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FitBuddy – AI Fitness Plan Generator using Gemini Models

A PROJECT REPORT

Submitted in partial fulfilment of the requirements
for the degree of

BACHELOR OF COMPUTER SCIENCE

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PROJECT TITLE: FitBuddy – AI Fitness Plan Generator using Gemini Models

1. ABSTRACT

FitBuddy is an AI-powered fitness application that generates personalized workout and diet plans. Traditional fitness apps provide same plan for everyone, but FitBuddy uses Google's Gemini Models (Gemini 1.5 Flash) to create a unique plan based on user's Age, Gender, Height, Weight, BMI, Goal, Fitness Level, and Diet preference. The system uses prompt engineering to interact with Gemini LLM and gives a structured, Indian-context diet and workout plan. This project aims to make fitness guidance affordable and accessible for students and beginners.

2. INTRODUCTION

Nowadays everyone wants to be fit, but personal trainer fees are high. YouTube videos are generic and not suitable for everyone. With AI, we can create a virtual personal trainer. FitBuddy solves this by taking user details and generating a customized plan in 5 seconds using Generative AI. It considers Indian foods like Chapati, Curd Rice, Sambar which other foreign apps don't give.

3. PROBLEM STATEMENT

- Existing fitness apps provide generic plans.
- No consideration for user's BMI, medical condition, veg/non-veg.
- No Indian diet options.
- Personal trainer is costly.

4. OBJECTIVES

1. To generate personalized fitness plan automatically.
2. To calculate BMI and give analysis.
3. To provide Indian-based diet plan.

4. To use Gemini AI for intelligent plan generation.
5. To make it user-friendly for college students.

5. EXISTING SYSTEM vs PROPOSED SYSTEM

Existing System: MyFitnessPal, HealthifyMe - they give pre-defined templates, less AI, subscription needed.

Proposed System: FitBuddy - Fully AI driven, free Gemini API, personalized table format output, considers medical issues and diet type, instant generation.

6. SYSTEM ARCHITECTURE

Architecture 4 layer la irukku:

Layer 1: User Input Layer (Frontend - Streamlit)

User Age, Height, Weight, Goal, Fitness Level, Days, Diet, Medical Issue enter pannuvan.

Layer 2: Processing Layer

BMI = Weight / (Height in meter)² nu calculate pannum. Eg: 70kg / (1.75*1.75) = 22.8

Layer 3: AI Engine Layer (Gemini Model)

We use `gemini-1.5-flash` model.

Prompt Engineering: We send a structured prompt with all user details to Gemini API. Gemini returns structured workout + diet plan.

Why Gemini?

- 1.5 Flash is Fast and Free tier irukku
- Multimodal & good reasoning
- Better than GPT for structured table output
- Context window large (1 Million tokens)

Layer 4: Output Layer

Result ah Table format la display pannum - Weekly Workout Table, Daily Diet Table, Tips, Motivational Quote.

Workflow: User Input -> BMI Calculation -> Prompt Creation -> Gemini API Call -> Response Parsing -> Display Plan

7. MODULES

1. User Data Collection Module: Collects all personal details through UI.

2. BMI Analyzer Module: Calculates BMI and classifies as Underweight (<18.5), Normal (18.5-24.9), Overweight (25-29.9), Obese (>30).

3. Prompt Engineering Module: This is heart of project. We create a dynamic prompt:

> "You are an expert fitness trainer... User age X, goal Y... Create table..."

4. AI Generation Module: Connects to Google Generative AI using API Key and generates plan.

5. Plan Display Module: Shows workout plan (Day wise: Chest, Back, Legs... with Sets x Reps) and diet plan (Breakfast, Lunch, Snack, Dinner with Calories).

8. TECHNOLOGY STACK

- *Language:* Python

- *Frontend:* Streamlit

- *AI Model:* Google Gemini 1.5 Flash (google-generativeai library)

- *Logic:* Python, Prompt Engineering

9. ADVANTAGES

1. 100% Personalized Plan

2. Free of Cost (Gemini free API)

3. Indian Food Diet Support

4. Instant Result - 5 seconds la

5. Beginner Friendly

6. Can be extended to WhatsApp Bot / Mobile App

10. FUTURE SCOPE

- Add Progress Tracking graph
- Add Exercise Video demonstration using Gemini Vision
- Add Voice Assistant like Alexa
- Integrate with Smartwatch for calorie tracking
- Convert to Android App

11. CONCLUSION

FitBuddy project successfully demonstrates how Generative AI (Gemini) can be used in health and fitness field. It replaces expensive personal trainers and gives scientifically structured plans. This project is useful for final year college demo and has high real-world startup potential.
